

Cluster Robust Variance Estimation

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Introduction

- ▶ The variance of the sum of 2 random variables X_1 and X_2 is:

$$\text{Var}(X_1) + \text{Var}(X_2) + 2 \cdot \underbrace{\text{cov}(X_1, X_2)}_{=0 \text{ if } X_1 \perp X_2}$$

- ▶ The OLS estimator can be written as the true parameter plus a sample (weighted) average of individual scores

$$\hat{\beta}_{\text{ols}} = \beta + \frac{\sum_i X_i u_i}{\sum_i X_i^2}$$

Asymptotic

- ▶ To conduct hypothesis tests we need the asymptotic distribution of our estimators
- ▶ I.e. we need to understand how the estimator would behave under repeated sampling.

$$\sqrt{N} \left(\hat{\beta}_{\text{ols}} - \beta \right) = \frac{\frac{1}{\sqrt{N}} \sum_i X_i u_i}{\frac{1}{N} \sum_i X_i^2}$$

- ▶ Focus on the numerator

$$\frac{1}{\sqrt{N}} \sum_i X_i u_i$$

- ▶ This is a sum of N random variables
- ▶ If they are all independent:

$$\xrightarrow{d} \mathcal{N} \left(0, \frac{1}{N} \sum_i \text{Var}(X_i u_i) \right)$$

Which is the problem?

- ▶ Inside a cluster the scores of 2 individuals are not independent
- ▶ Usual White Variance estimator is not consistent

$$\frac{1}{\sqrt{N}} \sum_{i=1}^N X_i u_i \xrightarrow{d} \mathcal{N} \left(0, \frac{1}{N} \sum_{i=1}^N \text{Var}(X_i u_i) + \frac{2}{N} \sum_{i \neq j}^N \sum_{j=1}^N \text{cov}(X_i u_i, X_j u_j) \right)$$

Literature review

Why Cluster?

- ▶ Traditional view: adjust for **within-cluster error correlation**.
- ▶ Bias in OLS S.E. grows with: strong error correlation, correlated regressors, large clusters.
- ▶ CRVE relies on **large G** asymptotics.

Fixed G

- ▶ Ibragimov and Müller (2010): inference can use **cluster-level estimates**.
- ▶ If they are (approximately) independent & Gaussian a **t-test** with t_{G-1} is valid even with heterogeneous variances.

The Design-Based Argument

- ▶ Clustering matters because of **sampling or assignment design**, not residual correlation.
- ▶ Residual correlation is **neither necessary nor sufficient**.
- ▶ If neither sampling nor treatment assignment is clustered **do not cluster**.
- ▶ CRVE is often conservative unless effects are homogeneous or clusters are rarely observed.

Heterogeneous effect

$$y_{ig} = \beta_g \mathbf{X}_{ig} + \gamma \mathbf{Z}_g + u_{ig}$$

$$\beta_g = \begin{bmatrix} \alpha_g \\ \beta_g \end{bmatrix}$$

Estimand

- ▶ In which parameter are we interested?



β_{APE}

Average Partial Effect

Ibragimov, Rustam, and Ulrich K Müller. 2010. "T-Statistic Based Correlation and Heterogeneity Robust Inference." *Journal of Business & Economic Statistics* 28 (4): 453–68.